Problem 3.12

An investor purchases \$1000 worth of units in a mutual fund whose units are valued at 4.00. The investment dealer takes a 9% "front-end load" from the gross payment. One year later the units have a value of 5.00 and the fund managers claim that the "fund's unit value has experienced a 25% growth in the past year." When units of the fund are sold by an investor, there is a redemption fee of 1.5% of the value of the units redeemed.

(a) If the investor sells all his units after one year, what is the effective annual rate of interest of his investment?

(b) Suppose instead that after one year the units are valued at 3.75. What is the return in this case?

Solution

The initial gross payment is \$1000. Since there is a 9% front-end load, the amount actually invested is

$$1000(1 - 0.09) = 910.$$

Since the unit price at purchase is 4.00, the number of units purchased is

$$\frac{910}{4} = 227.5.$$

(a) After one year, the units are worth 5.00 each, so the value of the investment is

$$227.5 \times 5 = 1137.5.$$

The redemption fee is 1.5% of this amount:

$$1137.5 \times \frac{1.5}{100} = 17.0625.$$

Hence the investor actually receives

$$1137.5 - 17.0625 = 1120.4375.$$

Therefore, the effective annual rate of interest is

$$i = \frac{1120.4375 - 1000}{1000} = 0.1204375 \approx 12.04\%.$$

(b) If instead the unit value after one year is 3.75, then the value of the investment is

$$227.5 \times 3.75 = 853.125.$$

The redemption fee is

$$853.125 \times \frac{1.5}{100} = 12.796875.$$

So the investor receives

$$853.125 - 12.796875 = 840.328125.$$

Thus, the return is

$$i = \frac{840.328125 - 1000}{1000} = -0.159671875 \approx -15.96\%.$$